Eslam Osama Mahmoud

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Professional Experience

Zikola *Mansoura, Egypt*
AI Engineer *April 2026 – Present*

- **Multi-Modal Conversational AI Agent:** Engineered stateful agentic workflows using LangGraph and LangChain, integrating LLMs, a tailored RAG pipeline, and STT/TTS engines; integrated WhatsApp APIs and web platforms to deliver real-time, preference-based customer recommendations.
- **Real-Time Audio & Meeting Intelligence Engine:** Built dual-channel audio capture (tab + mic) via a Chrome MV3 Extension; implemented a custom Python word-alignment algorithm (±1.5s) with DOM metadata parsing to filter audio bleed and map Deepgram Nova-3 STT diarization tags to attendee names.
- **Async LLM Processing Pipeline:** Architected a parallel `asyncio` pipeline to split multi-hour transcripts into dynamic chunks, featuring custom prompt engineering and a state-machine auto-repair parser guaranteeing valid JSON outputs for meeting action items, summaries, and sentiment scores.
- **Enterprise Security & RBAC Architecture:** Designed passwordless authentication using WebAuthn Passkeys and SHA-256 OTPs; enforced multi-tenant RBAC with automated offboarding services, JWT invalidation, and immutable audit logging.
- **High-Throughput Streaming & Optimization:** Overcame Vercel's 4.5MB payload constraint using 3MB chunked BytesIO streaming; persisted hybrid transcript analytics in PostgreSQL via SQLAlchemy for dashboard re-analysis.
- **Multi-Platform Media Analytics & Automation:** Developed an automated media-buying ETL engine processing multi-platform CSVs (Meta, TikTok, Snapchat) with auto-calculated KPIs (ROAS, CPA, CTR, CPC) integrated into n8n AI-agent workflows.

Projects

Elara – AI-Powered Educational Ecosystem *Multi-Module System*

- **Elara TTS (Egyptian Arabic Engine)** GitHub
  Curated a 14-hr Egyptian speech dataset (∼18.9k samples, VAD via faster-whisper) with custom NLP normalization; fine-tuned Spark-TTS (0.5B) via Unsloth and Wav2Vec2/BiCodec tokenization for educational speech.
- **LLM Cognitive Evaluation Engine** GitHub
  Fine-tuned Qwen2.5-14B via QLoRA (4-bit NF4) on 3,946 sessions; deployed on Modal with vLLM + guided JSON decoding to auto-generate schema-validated reports on student misconceptions.
- **RAG Learning Assistant** GitHub
  Built an ingestion pipeline with recursive chunking and FAISS vector retrieval to ground LLM responses in course materials with precise source citations.
- **Student KPIs & Predictive Analytics** GitHub
  Developed end-to-end ML pipelines using gradient-boosted ensembles to forecast academic performance and feed real-time instructor monitoring dashboards.

LUNA16 3D Pulmonary Nodule Detection System GitHub

- Built a 3-stage 3D pipeline on 888 CT scans with classical 7-step lung segmentation in NumPy/SciPy (HU thresholding, morphological closing) and trained DenseNet-121 from scratch for high-recall candidate detection and false positive reduction.

Arabic Medical LLM Fine-Tuning & Deployment GitHub

- Fine-tuned Qwen2.5-1.5B via LoRA/Unsloth on merged Arabic medical QA datasets; deployed an async CUDA inference FastAPI backend with responsive chat UI.

Building Energy Consumption Digital Twin GitHub

- Hierarchical regression system combining XGBoost sub-models with a Neural Network meta-learner, served via Flask REST API with real-time scenario simulation.

Technical Skills

Programming	Python, C, C++, PHP, JavaScript, TypeScript
AI & ML Frameworks	PyTorch, TensorFlow, Scikit-learn, Keras, XGBoost
Data Science	Pandas, NumPy, Matplotlib, Seaborn, OpenCV
NLP & Speech	Hugging Face Transformers, PEFT/TRL, NLTK, Wav2Vec2, Faster-Whisper
MLOps & Deployment	Docker, FastAPI, Flask, Streamlit, vLLM, Unsloth, Vercel Serverless
Web & Integration	React, MV3, n8n, Google Apps Script, WebAuthn, Webhooks
Databases & Tools	PostgreSQL, SQL Server, SQLAlchemy, Git, GitHub, Jupyter Notebook
Languages	Arabic (Native), English (Fluent)

Education

Bachelor of Computer Science *2022 – 2026*
Faculty of Computers and Information, Mansoura University, Egypt *Graduated*

Summer Training

- **Data Analysis & Freelancing** – ITIDA & NTI *120 hrs, Aug–Sep 2024*
  Covered data analytics workflow, freelancing dynamics, proposal writing, and project pricing strategies.
- **Data Science & Web Development** – MCIT *120 hrs, Jul–Aug 2025*
  Hands-on training in core data science concepts, business domain modeling, and web development.